

Unit 5: Square Root Functions

Practice Worksheet

Name: _____

Date: _____

Period: _____

Section 1: Simplifying Radicals

Simplify each radical expression.

1. $\sqrt{98}$

2. $\sqrt{45x^6}$

3. $\sqrt[3]{40y^7}$

4. $\sqrt{72m^5n^8}$

5. $\sqrt[3]{128a^6b^4}$

Section 2: Radical $\leftrightarrow$ Rational Exponents

Convert to rational exponent form.

1. $\sqrt{x^3}$

2. $\sqrt[3]{y^4}$

3. $\sqrt[5]{m^2}$

4. $(\sqrt[4]{n})^5$

Convert to radical form.

5. $x^{2/5}$

6. $y^{7/3}$

7. $a^{1/4}$

8. $b^{5/2}$

Section 3: Domain and Range

Find the domain and range of each function.

1. $f(x) = \sqrt{x-4} + 2$

2. $f(x) = -3\sqrt{x+1} - 5$

3. $f(x) = \frac{1}{2}\sqrt{x+7} + 3$

4. $f(x) = -\sqrt{x-2}$

Section 4: Transformations

Describe ALL transformations from $y = \sqrt{x}$.

1. $f(x) = \sqrt{-x-5} - 3$

2. $f(x) = -2\sqrt{x-4}$

3. $f(x) = \frac{1}{3}\sqrt{2x-6} + 1$

4. $f(x) = 5\sqrt{x+2} - 8$

Section 5: Identifying Endpoints

Find the endpoint of each function.

1. $f(x) = \sqrt{x-7} + 2$

2. $f(x) = -\sqrt{x+3} - 4$

3. $f(x) = 2\sqrt{x-1}$

4. $f(x) = \frac{1}{4}\sqrt{-x-8} - 5$

Section 6: Composition to Check Inverses

Determine if each pair of functions are inverses. Show $f(g(x))$ and $g(f(x))$.

1. $f(x) = x^2 - 3$ and $g(x) = \sqrt{x + 3}$

3. $f(x) = x^2 + 5$ and $g(x) = -\sqrt{x - 5}$

2. $f(x) = (x - 1)^2 + 4$ and $g(x) = \sqrt{x - 4} + 1$

Section 7: Finding the Inverse

Find the inverse of each function.

1. $f(x) = \sqrt{x - 3} + 1$

3. $f(x) = \sqrt{x - 1} + 6$

2. $f(x) = \sqrt{x + 5} - 2$

Section 8: Solving Square Root Equations

Solve each equation. Check for extraneous solutions!

1. $\sqrt{3x-2} = 4$

4. $\sqrt{x+3} = x+1$

2. $\sqrt{x+5} - 3 = 1$

5. $2(x+1)^{1/2} - 6 = 0$

3. $\sqrt{2x-1} + 2 = 7$

6. $\sqrt{4-x} - 2 = x$

Section 9: Writing Equations from Information

Write the equation for each square root function.

1. Endpoint: $(3, -2)$, Point: $(7, 2)$

3. Endpoint: $(5, 0)$, Point: $(9, 6)$

2. Endpoint: $(-1, 4)$, Point: $(8, 1)$

ANSWER KEY

Section 1: Simplifying Radicals

- $7\sqrt{2}$
- $3x^3\sqrt{5}$
- $2y^2\sqrt[3]{5y}$
- $6m^2n^4\sqrt{2m}$
- $4a^2b\sqrt[3]{2b}$

Section 2: Rational Exponents

To Rational Exponent Form:

- $x^{3/2}$
- $y^{4/3}$
- $m^{2/5}$
- $n^{5/4}$

To Radical Form:

- $\sqrt[5]{x^2}$
- $\sqrt[3]{y^7}$ or $y^2\sqrt[3]{y}$
- $\sqrt[4]{a}$
- $\sqrt{b^5}$ or $b^2\sqrt{b}$

Section 3: Domain and Range

- $D : x \geq 4, \quad R : y \geq 2$
- $D : x \geq -1, \quad R : y \leq -5$
- $D : x \geq -7, \quad R : y \geq 3$
- $D : x \geq 2, \quad R : y \leq 0$

Section 4: Transformations

- Reflect y -axis, Left 5, Down 3
- Right 4, Reflect x -axis, Vert. stretch by 2
- Right 3, Up 1, Horiz. comp. by 2, Vert. comp. by $\frac{1}{3}$
- Left 2, Down 8, Vert. stretch by 5

Section 5: Endpoints

- $(7, 2)$
- $(-3, -4)$
- $(1, 0)$
- $(-8, -5)$

Section 6: Composition/Inverses

- $f(g(x)) = x, g(f(x)) = |x|$
NOT inverses (only for $x \geq 0$)
- $f(g(x)) = x, g(f(x)) = x$
YES, inverses (for $x \geq 1$)
- $f(g(x)) = x, g(f(x)) = -|x|$
NOT inverses

Section 7: Finding the Inverse

- $f^{-1}(x) = (x - 1)^2 + 3$
or $x^2 - 2x + 4$
- $f^{-1}(x) = (x + 2)^2 - 5$
or $x^2 + 4x - 1$
- $f^{-1}(x) = (x - 6)^2 + 1$
or $x^2 - 12x + 37$

Section 8: Solving Equations

- $x = 6$
- $x = 11$
- $x = 13$
- $x = 1$ only ($x = -2$ is extraneous)
- $x = 8$
- $x = 0$ only ($x = -5$ is extraneous)

Section 9: Writing Equations

- $f(x) = 2\sqrt{x - 3} - 2$
- $f(x) = -\sqrt{x + 1} + 4$
- $f(x) = 3\sqrt{x - 5}$